# A SPACE THAT BECOMES A SCHEME AFTER A PURELY INSEPARABLE BASE CHANGE IS A SCHEME

THE STACKS PROJECT AUTHORS

*Editorial note.* Counted span is the author's own proof environment, source lines 1801-1806, immediately following the statement at 1794-1799.

*Editorial note.* One cross-chapter pointer (Limits of Spaces) is recorded in assembly.external_references.

*Editorial note.* Hypothesis-verification pattern: the span's own content is the transfer of 'purely inseparable' to the descent lemma's three adjectives.

*Allowed standard prerequisites (editorial).*

- Descent of scheme-ness along integral universally injective surjective morphisms. Limits of Spaces, Lemma spaces-limits-lemma-integral-universally-bijective-scheme, cited at proof lines 1803-1805. Terminal input: X is a scheme once X -> X_{k'} is integral, surjective, and universally injective and X_{k'} is a scheme.

*Source reference.* Original reference 07VV: focused-seeds/geometry/S9-sof-scheme-after-purely-inseparable-base-change/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $k$ be a field. Let $X$ be an algebraic space over $k$. If there exists a purely inseparable field extension $k'/k$ such that $X_{k'}$ is a scheme, then $X$ is a scheme.*

*Proof.* The morphism $X_{k'} \to X$ is integral, surjective, and universally injective. Hence this lemma follows from Limits of Spaces, Lemma 07VV. □

*Date*: stacks-project:spaces-over-fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.